

Computer Network

MCA III Semester

MCA 5122

A BRIEF HISTORY

- A *network* is a group of connected, communicating devices such as computers and printers.
- An *internet* is two or more networks that can communicate with each other.
- The most notable internet is called *The Internet*, composed of hundreds of thousands of interconnected networks.

A BRIEF HISTORY— ARPANET

- The **A**dvanced **R**esearch **P**rojects **A**gency (ARPA) in the Department of Defense (**D**o**D**) was interested in finding a way to connect computers together so that the researchers could share their findings.
- In 1967, at an **A**ssociation for **C**omputing **M**achinery (ACM) meeting, **ARPA** presented its ideas for **ARPANET**, a small network of connected computers. **Host** is **connected to** a specialized computer, called an **I**nterface **M**essage **P**rocessor (**IMP**) & IMPs are connected to communicate to each other.

A BRIEF HISTORY

- By 1969, ARPANET was a reality.
- Four nodes, at the University of California at Los Angeles (UCLA), the University of California at Santa Barbara (UCSB), Stanford Research Institute (SRI), and the University of Utah, were connected via the IMPs to form a Network Software called the Network Control Protocol (NCP) provided communication between the hosts.

A BRIEF HISTORY - Birth of the Internet

- In 1972, Vint Cerf and Bob Kahn, both of whom were part of the core ARPANET group, collaborated on what they called the Internetting Project.
- The idea was to link different networks together so that a host on one network could communicate with a host on a second, different network.
 - Challenges – varying size of packets , transmission rate & varying standards etc. at different networks.
 - Cerf and Kahn devised the idea of a device called a gateway to serve as the intermediary hardware to transfer data from one network to another.

A BRIEF HISTORY- TCP

- Cerf and Kahn's landmark 1973 paper outlined the protocols to achieve end-to-end delivery of data. This was a new version of NCP (TCP).
- In October 1977, an internet consisting of three different networks (ARPANET, packet radio, and packet satellite) was successfully demonstrated.

A BRIEF HISTORY- TCP/IP

- Authorities made a decision to **split TCP** into two protocols: Transmission Control Protocol (**TCP**) and Internet Protocol (**IP**).
 - **IP** would handle datagram **routing**.
 - **TCP** would be responsible for **higher level functions** such as **segmentation, reassembly, and error detection**.
- The new combination became known as **TCP/IP**.

A BRIEF HISTORY

- In 1981, under a DARPA contract, UC Berkeley modified the UNIX operating system to include TCP/IP. This inclusion of network software along with a popular operating system did much for the popularity of networking.
- In 1983, authorities abolished the original ARPANET protocols, and TCP/IP became the official protocol. Those who wanted to use the Internet to access a computer on a different network had to be running TCP/IP.

PROTOCOLS

A protocol is a set of rules that governs communication.-

For example, in a telephone conversation, there are a set of rules that we need to follow. There is a rule how to make connection (**dialing the telephone number**), how to respond to the call (**picking up the receiver**), how to greet, how to let the communication flow smoothly by listening when the other party is talking, and finally how to end the communication (**hanging up**)

For communication to occur, the entities must agree on a protocol. A protocol defines **what** is communicated, **how** it is communicated, and **when** it is communicated. The key elements of a protocol are **syntax**, **semantics**, and **timing**.

Syntax. Syntax refers to the structure or format of the data, meaning the order in which they are presented. For example, a simple protocol might expect the first 8 bits of data to be the address of the sender, the second 8 bits to be the address of the receiver etc.

Semantics. Semantics refers to the meaning of each section of bits. How is a particular pattern to be interpreted.

Timing. Timing refers to two characteristics: **when data should be sent** and **how fast it can be sent**. Order of messages (TCP Three-Way Handshake-order SYN, SYN+ACK, ACK), Synchronization, Acknowledgement Timeout

STANDARDS ORGANIZATION

Standards are developed through the cooperation of -

- ✓ Standards Creation Committees
- ✓ Forums
- ✓ Regulatory Agencies

Why Standards ?

Standards are essential in creating and maintaining an **open and competitive market** for equipment manufacturers and also in **guaranteeing national and international interoperability** of data and telecommunications technology and processes.

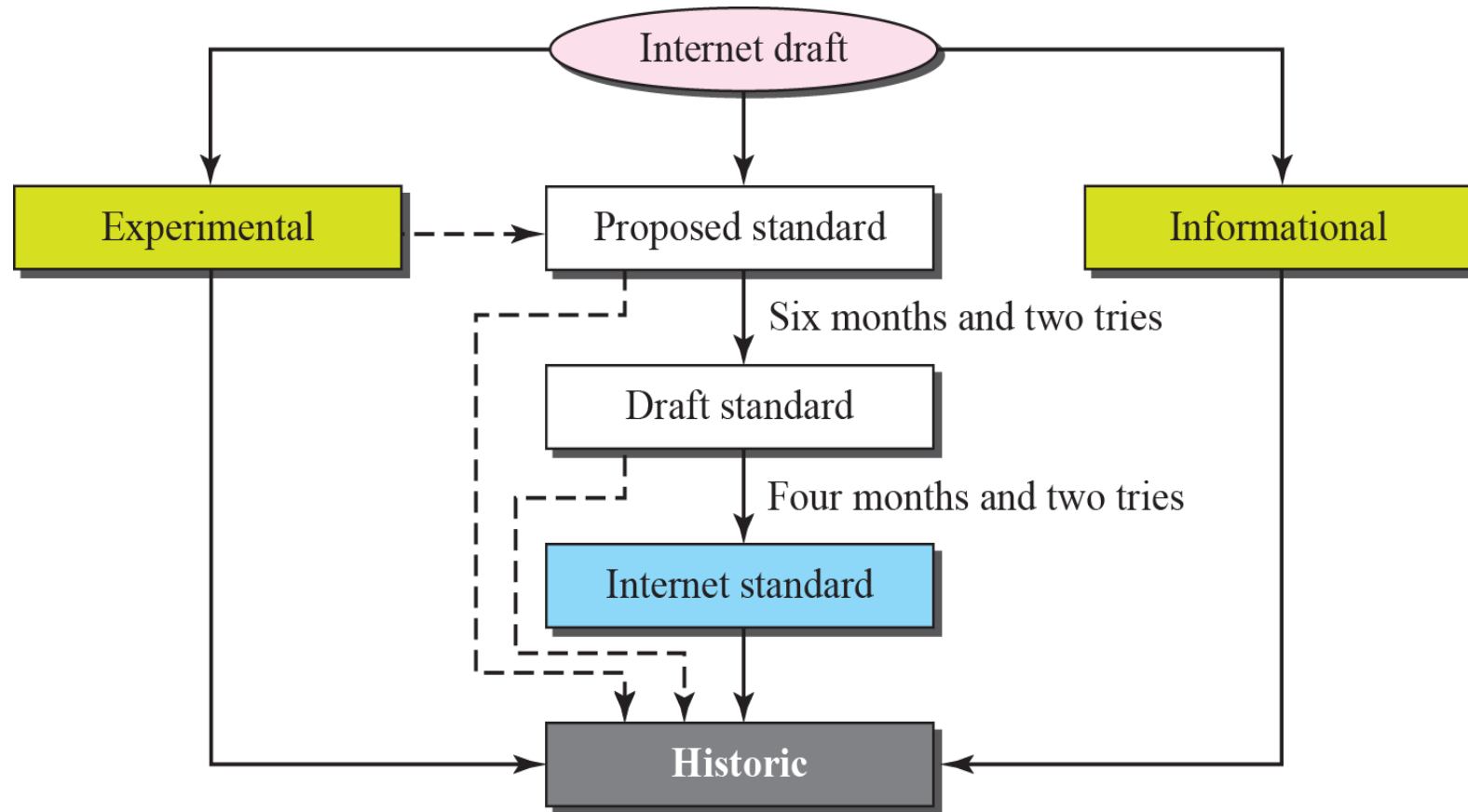
De facto. Standards that have not been approved by an organized body but have been adopted as standards through widespread use are de facto standards.

De jure. De jure standards are those that have been legislated by an officially recognized body.

INTERNET STANDARDS

- An Internet standard is a thoroughly **tested specification** that is useful to and adhered to by those who work with the Internet.
- A specification begins as an **Internet draft**.
- An Internet draft is a working document with **no official status** and a **six-month lifetime**.
- Internet authorities, a **draft may be published** as a **Request for Comment (RFC)**.
 - Each RFC is edited, **assigned a number**, and **made available to all interested parties**.
 - RFCs go **through maturity levels** and are categorized according to their requirement level.

Figure 1.2 *Maturity levels of an RFC*



An RFC, during its lifetime, falls into one of six **maturity levels**: proposed standard, draft standard, Internet standard, historic, experimental, and informational .

*RFCs can be found at
<http://www.rfc-editor.org>.*



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The Series

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Mailing Lists

rfc-dist@rfc-editor.org is for RFC publication announcements. rfc-interest@rfc-editor.org is for discussion of the RFC series and related functions.

The RFC Series

contains technical and organizational documents about the Internet, including the specifications and policy documents produced by four streams: the Internet Engineering Task Force (IETF), the Internet Research Task Force (IRTF), the Internet Architecture Board (IAB), and Independent Submissions.

Browse the RFC Index

HTML (ascending) • HTML (descending) • TXT • XML
Note: These files are large.

Browse RFCs by Status

Internet Standard

Draft Standard • Proposed Standard

Best Current Practice

Informational • Experimental • Historic

Uncategorized (Early RFCs)

.....

Official Internet Protocol Standards

RFC Status Changes

See you at IETF 114!

Office Hours are here

Recent RFCs

- RFC 9252: BGP Overlay Services Base Segment Routing over IPv6 (SRv6)
- RFC 9254: Encoding of Data Modeled YANG in the Concise Binary Object Representation (CBOR)
- RFC 9241: Content Delivery Network Interconnection (CDNI) Footprint and Capabilities Advertisement Using App Layer Traffic Optimization (ALTO)
- RFC 9240: An Extension for Application Traffic Optimization (ALTO): Entity Pro Maps
- RFC 9261: Exported Authenticators in
- RFC 9231: Additional XML Security Un Resource Identifiers (URIs)
- RFC 9249: A YANG Data Model for NTI
- RFC 9280: RFC Editor Model (Version 3)
- RFC 9281: Entities Involved in the IETF Standards Process
- RFC 9282: Responsibility Change for the Series

Sponsor



Requirement Levels

RFCs are classified into five **requirement levels**: required, recommended, elective, limited use, and not recommended (see Figure)

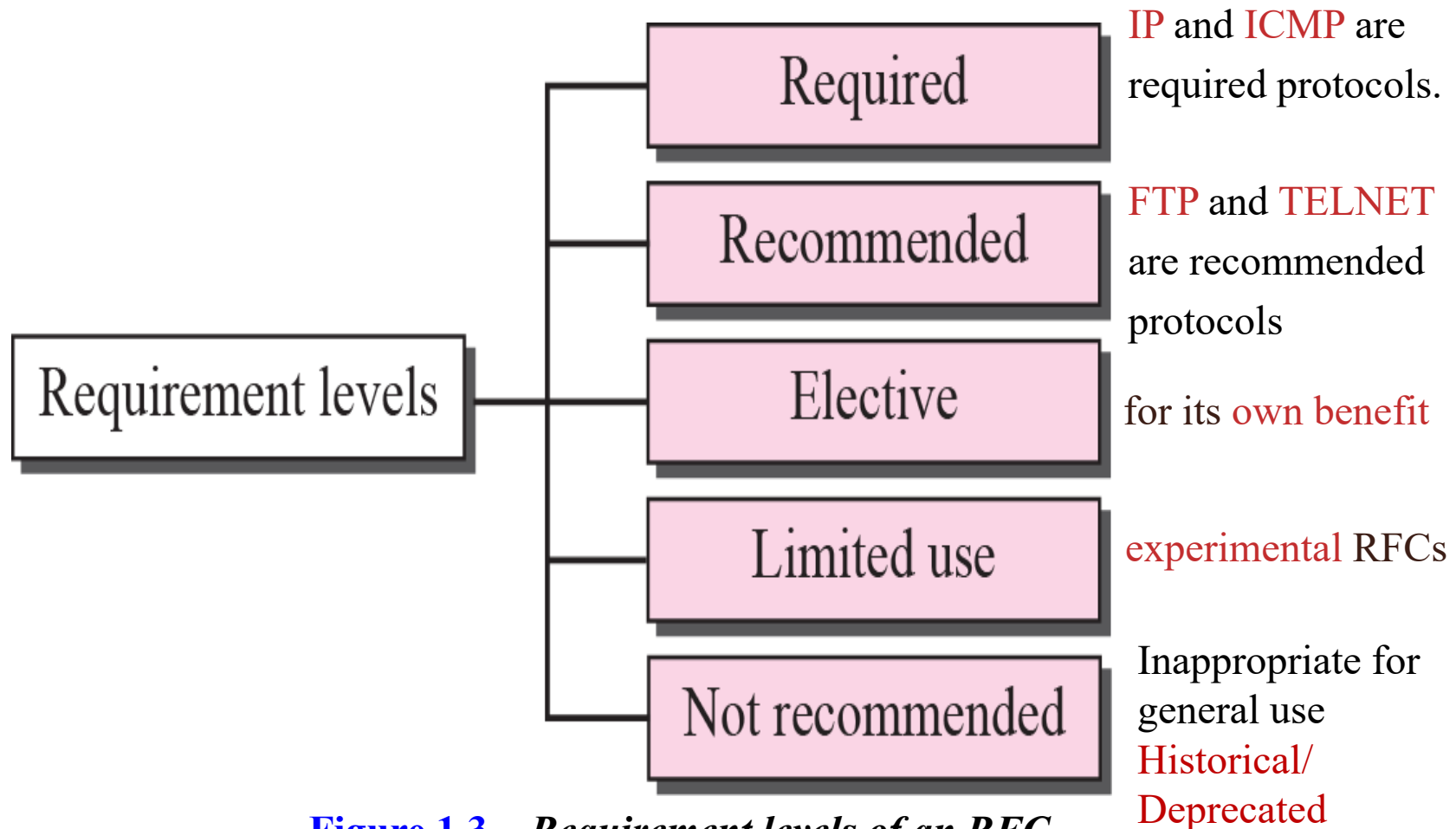


Figure 1.3 *Requirement levels of an RFC*

Some Required and Recommended Protocols

6. The Protocols

6.1. Standard Protocols

Protocol	Name	Status	RFC
-----	----	-----	---
	Assigned Numbers	Required	1010
	Gateway Requirements	Required	1009
IP	Internet Protocol	Required	791
	as amended by:		
	IP Subnet Extension	Required	950
	IP Broadcast Datagrams	Required	919
	IP Broadcast Datagrams with Subnets	Required	922
ICMP	Internet Control Message Protocol	Required	792
UDP	User Datagram Protocol	Recommended	768
TCP	Transmission Control Protocol	Recommended	793
DOMAINS	Domain Name System	Recommended	1034,1035
TELNET	Telnet Protocol	Recommended	854
FTP	File Transfer Protocol	Recommended	959
SMTP	Simple Mail Transfer Protocol	Recommended	821

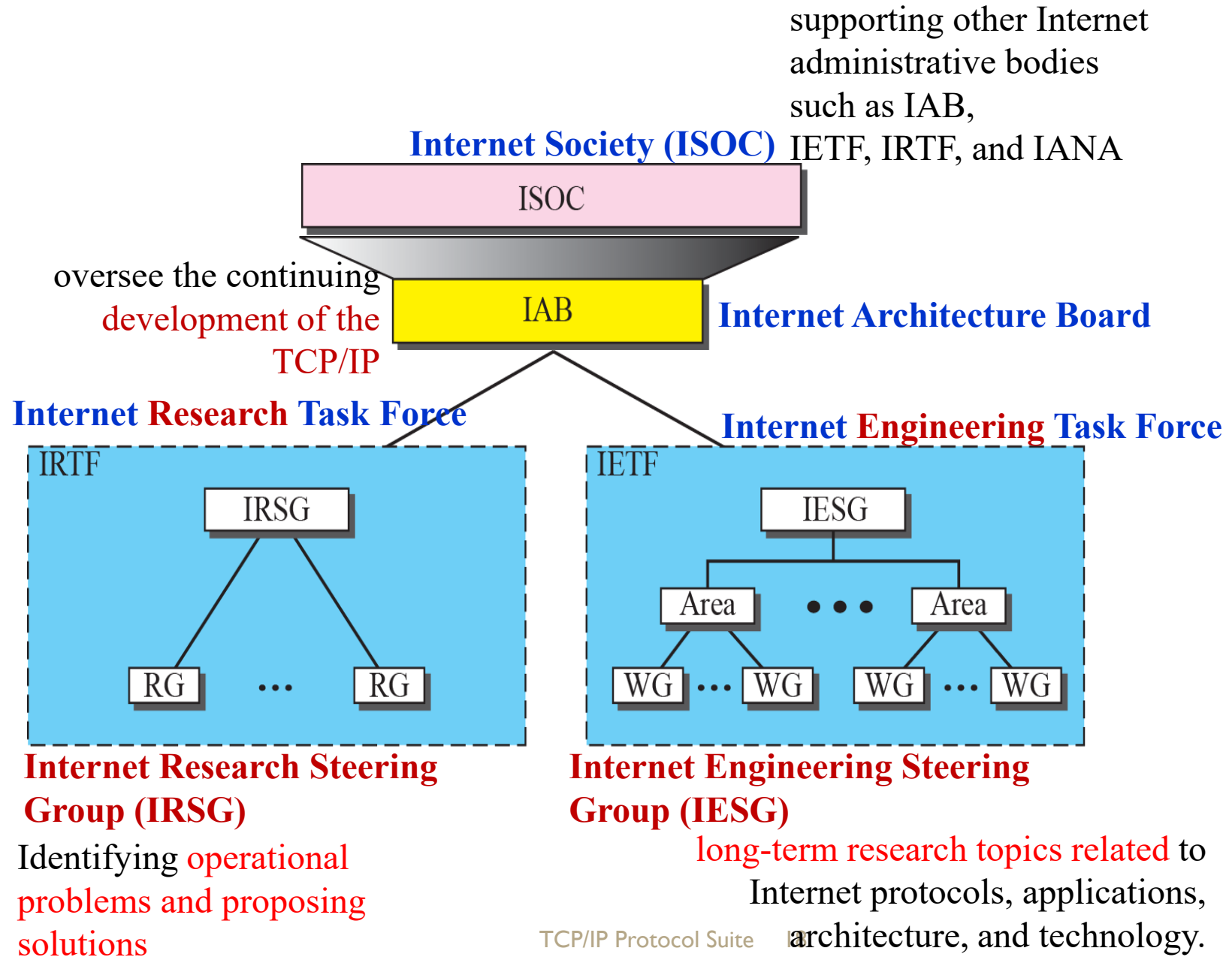
Some Elective Protocols and RFCs

STATSRV	Statistics Server	Elective	996
NNTP	Network News Transfer Protocol	Elective	977
NICNAME	WhoIs Protocol	Elective	954
HOSTNAME	HOSTNAME Protocol	Elective	953
POP2	Post Office Protocol	Elective	937
SFTP	Simple File Transfer Protocol	Elective	913
RLP	Resource Location Protocol	Elective	887
RTELNET	Remote Telnet Service	Elective	818
TFTP	Trivial File Transfer Protocol	Elective	783
FINGER	Finger Protocol	Elective	742
SUPDUP	SUPDUP Protocol	Elective	734
NETED	Network Standard Text Editor	Elective	569
RJE	Remote Job Entry	Elective	407

Some Not Recommended Protocols

Protocol	Name	Status	RFC
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SGMP	Simple Gateway Monitoring Protocol	Not Recommended	1028
HMP	Host Monitoring Protocol	Not Recommended	869
GGP	Gateway Gateway Protocol	Not Recommended	823
CLOCK	DCNET Time Server Protocol	Not Recommended	778
MPM	Internet Message Protocol	Not Recommended	759
NETRJS	Remote Job Service	Elective	740
XNET	Cross Net Debugger	Elective	IEN 158
NAMESERVER	Host Name Server Protocol	Not Recommended	IEN 116
MUX	Multiplexing Protocol	Not Recommended	IEN 90
GRAPHICS	Graphics Protocol	Not Recommended	NIC 24308

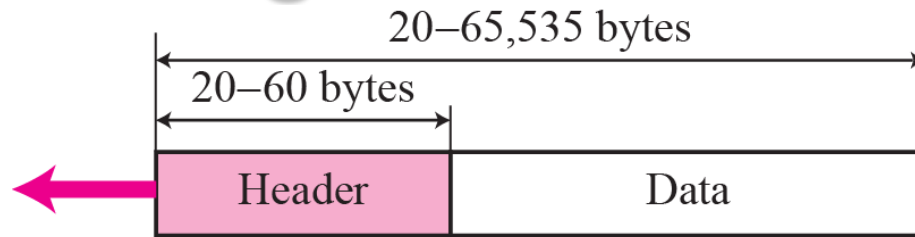
Figure 1.4 *Internet administration*



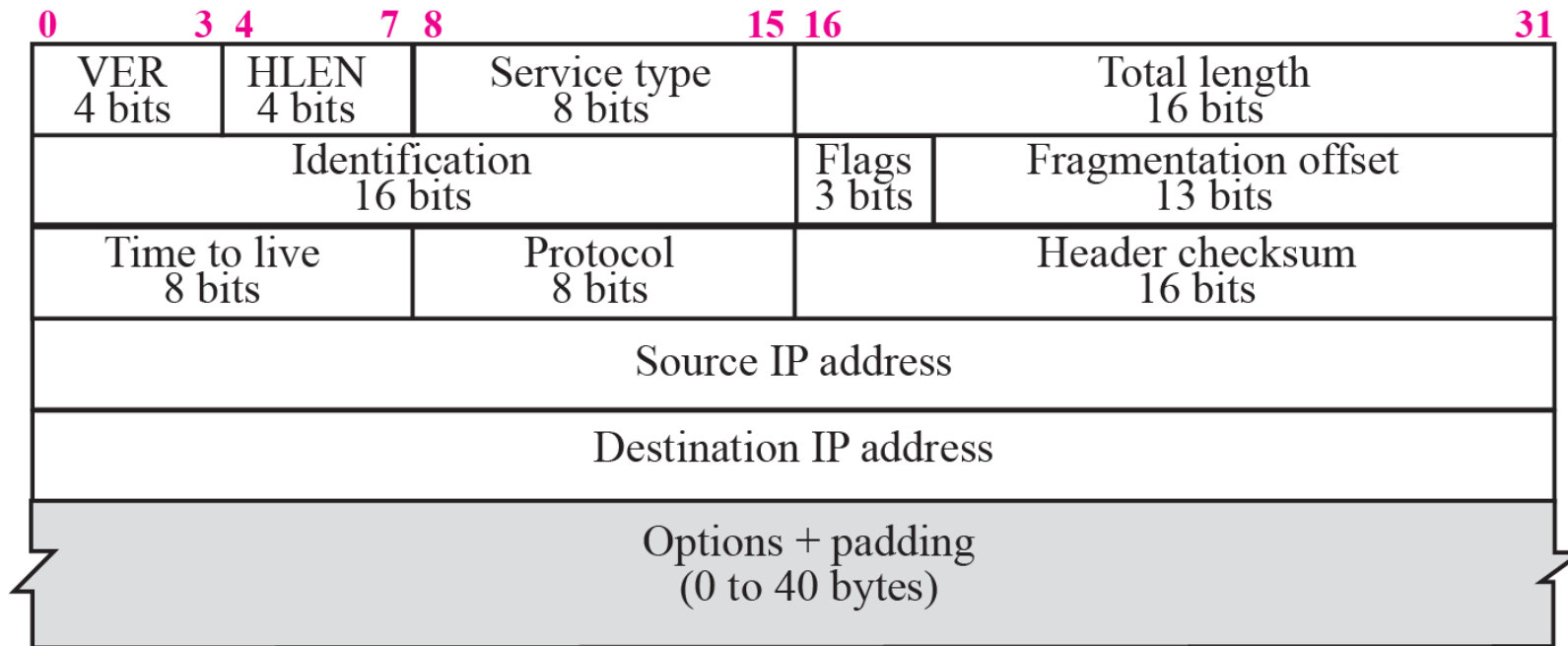
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IP Datagram



a. IP datagram



b. Header format